

# Introduction to Neurology

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## Introduction to Neurology / Investigations of the Central Nervous System

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### Objectives

1. To know the **clinical approach to neurological cases** via history taking and physical examination
2. To know the **relevant investigations**

### Outline

Introduction · Anatomy of the nervous system · History · Physical examination · Special diagnostic procedures · Relevant investigations · Further reading

### Introduction

**Neurology** is the branch of medicine concerned with the **study and treatment of disorders of the nervous system**.

The **nervous system** is a complex, sophisticated system that **regulates and coordinates body activities**.

It comprises:

- The **central nervous system** (brain and spinal cord)
- The **peripheral nerves**

The **brain** is the most complicated and intricate system that exists. It **processes and creates almost every aspect of our conscious experience**.

### Anatomy of the nervous system

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The brain has **4 regions**:

1. **Cerebrum**
2. **Diencephalon** (thalamus, hypothalamus, subthalamus, epithalamus)
3. **Brainstem** (midbrain, pons & medulla)
4. **Cerebellum**

## Cerebrum

The **cerebral hemispheres constitute the greatest mass of brain tissue.**

The outer layers are formed by the **cellular grey matter known as the cerebral cortex.**

**Consciousness** depends on interaction between **intact cerebral hemispheres** and the **activating (or arousal) system that lies in the diencephalon and upper brainstem.**

**Either extensive disease of the cerebral cortex OR damage to the activating system may impair consciousness to the point of coma.**

## Cerebellum

Primarily concerned with **coordination and balance.** It **refines movements.**

## Brainstem

Continues down to the spinal cord, and is **crucial for vital functions – breathing, heart rate, blood pressure and sleep-wake cycles.**

It has **three sections: midbrain, pons and medulla.**

## Spinal cord

A **cylindrical mass of nervous tissue encased within the bony vertebral column.**

It contains **long tracts** that connect the brain with the peripheral nervous system.

It also **mediates reflex activity** – an involuntary bodily response involving **3 basic components: a receiving apparatus, a nerve centre, and a responding apparatus.**

## History taking in neurology

**A detailed history is the cornerstone of neurologic assessment.**

**Informants** should be the **parents, and older children (> 3-4 years) should be questioned.**

## The conventional order of clerking in medicine

1. Presenting complaint (c/o or PC)
2. History of the presenting complaint (HPC)
3. Past medical history
4. Past surgical history
5. Pregnancy / perinatal / postnatal history
6. Immunization
7. Developmental milestones
8. Family and social history
9. Drug history / allergy

### 1. Chief complaint

The history should begin with the **chief complaint and its significance in the context of normal development.**

## 2. History of the presenting complaint

Should provide a **chronologic outline of the patient's symptoms**, with attention paid to:

- **Location**
- **Quality**
- **Intensity**
- **Duration**
- **Associated features**
- **Alleviating or exacerbating factors**

## 3. Complete birth history

Particularly if a **congenital or perinatal disorder** is suspected.

**Pregnancy** – ask about common complications such as **pregnancy-induced hypertension, pre-eclampsia, gestational diabetes, vaginal bleeding, infections, and falls**.

Also inquire about:

- **Drug** (prescription, herbal, illicit) or **alcohol** use
- **Fetal movement** – reduced or absent might indicate **neuromuscular disorders**
- Any **abnormal ultrasound or amniocentesis** results

**Gestational age at birth and mode of delivery** – SVD, vacuum- or forceps-assisted, C/S (elective or emergency, and the indication).

The **presence or absence of fetal distress**, any **resuscitation**, **birth weight**, and complicated hospital stay.

If the infant had **jaundice**, it is important to determine **both the degree of jaundice and how it was managed**.

## 4. Developmental history

**A major component of the neurologic history.**

Careful evaluation of a child's **domains of development** – **social, cognitive, language and motor skills (fine and gross)** – is required to distinguish normal development from either an **isolated** or a **global** (i.e. in two or more domains) developmental delay.

**A static abnormality in development from birth** suggests a **congenital, intrauterine, or perinatal** cause.

**A loss of skills (regression) over time** strongly suggests an **underlying degenerative disease of the CNS**, such as an **inborn error of metabolism**.

### Screening scheme for developmental delay

| Age (mo) | Gross motor                 | Fine motor                | Social skills            | Language skills |
|----------|-----------------------------|---------------------------|--------------------------|-----------------|
| <b>3</b> | Supports weight on forearms | Opens hands spontaneously | Smiles appropriately     | Coos, laughs    |
| <b>6</b> | Sits momentarily            | Transfers objects         | Shows likes and dislikes | Babbles         |

| Age (mo)  | Gross motor                    | Fine motor                    | Social skills                | Language skills        |
|-----------|--------------------------------|-------------------------------|------------------------------|------------------------|
| <b>9</b>  | Pulls to stand                 | Pincer grasp                  | Plays pat-a-cake, peek-a-boo | Imitates sounds        |
| <b>12</b> | Walks with 1 hand held         | Releases an object on command | Comes when called            | 1-2 meaningful words   |
| <b>18</b> | Walks upstairs with assistance | Feeds self from spoon         | Mimics actions of others     | At least 6 words       |
| <b>24</b> | Runs                           | Builds a tower of 6 blocks    | Plays with others            | 2- to 3-word sentences |

## 5. Family and social history

**Family history** – history of **neurologic disease including developmental delay, and inherited disorders, for all first- and second-degree relatives.**

**Miscarriages or fetal deaths** (gender, gestational age at the time of demise).

**Ethnic backgrounds** – are parents related to each other?

The **incidence of metabolic and degenerative disorders of the CNS is increased significantly in children of consanguineous marriages.**

**Social history** should detail the child's **current living environment** and the child's **relationship with other family members.**

Any **recent stressors** – divorce, remarriage, birth of a sibling, or death of a loved one – because they can affect the child's behaviour.

**Academic and social performance**, paying particular attention to any **abrupt changes**. The latest report card can be assessed.

**Peer relationships** can be evaluated by having the child name his or her best friends.

Any child who is unable to name at least two or three playmates might have abnormal social development.

## 6. Review of systems

It is **essential** to perform a review of systems.

Abnormalities of the CNS often manifest with **vague, non-focal symptoms that may be misattributed to other organ systems** – e.g. **vomiting, constipation, urinary incontinence.**

### Physical examination

The examination should be conducted in a **non-threatening, child-friendly setting.**

The child should be **allowed to sit where the child is most comfortable**, whether on a parent's lap or on the floor of the examination room.

The physician should **approach the child slowly**, reserving any **invasive, painful, or discomforting tests for the end** of the examination (e.g. measurement of head circumference, gag reflex).

The more that the examination seems like a game, the more the child will cooperate.

### General observation

The child's:

- **Appearance**
- **Behaviour**
- **Mental state**
- **Abnormal posture or motor function**
- **Head** – size, shape, fontanels, OFC

**Examination begins during the interview.** Observe the child's appearance and movements, **dysmorphic facies**, abnormality of motor function such as **hemiparesis or gait disturbance**.

The child's behaviour while playing and interacting with parents may also be telling. **A normal child usually plays independently early in the visit but then engages in the interview process.**

### Size of the head

**Microcephaly** – a small head frequently reflects a **small brain**. Cause: **perinatal or postnatal insult to the brain**.

**Macrocephaly** – most commonly **familial**, but may be from a **disturbance of growth, chromosomal defect, storage disorder or hydrocephalus**.

### The shape of the head

**Craniosynostosis** – **premature closure of cranial sutures**, causing a variety of abnormal head shapes.

**Chronic subdural haemorrhages** – **square or box-like shape**, due to long-standing presence of fluid in the subdural space.

**Plagiocephaly** (flattening of the skull) – can be seen in normal infants, but may be particularly prominent in **hypotonic or weak infants**, who are less mobile.

### Fontanels

**Anterior fontanel** – **diamond-shaped**, at the junction of the **frontal and parietal bones**, measures **2 x 2 cm**, closes **9-18 months**.

**Posterior fontanel** – **triangular shape**, at the junction of the **parietal and occipital bones**, closes **6-8 weeks**.

**A bulging fontanel** indicates **increased ICP** – also vigorous crying in a normal infant.

**A very small or absent anterior fontanel at birth** might indicate **craniosynostosis or microcephaly**.

### Occipitofrontal circumference (OFC)

**OFC reflects brain growth.** It should be measured and documented.

**The average rate of head growth:**

**A healthy premature infant:** 0.5 cm in the first 2 weeks, 0.75 cm in the 3rd week, and 1.0 cm in the 4th week and every week thereafter until the 40th week of development.

**An average term infant:** measures **34-35 cm at birth**; increases by **2 cm/month for the 1st 3 months**, **1 cm/month for the 2nd 3 months**, **0.5 cm/month for the last 6 months**; thereafter **0.25 cm/month from 1-3 years**, then **0.5 cm/year up to 6 years**. After 6 years it ceases to be relevant.

### The sequence of central nervous system examination

1. **Higher function**
2. **Cranial nerves**
3. **Motor functions** – posture, nutrition of muscle, DTR, power, superficial reflexes
4. **Sensory** – light touch, pain, temperature, joint position, vibration, stereognosis, Romberg's sign
5. **Autonomic function** – rest and exercise pulse and BP
6. **Soft neurologic signs** – neck stiffness, Kernig's sign and Brudzinski's sign
7. **Cerebellar function** – intention tremor, nystagmus, dysdiadochokinesia
8. **Gait** – hemiparetic, ataxia (cerebellar and sensory), spastic, steppage, myopathic, or waddling

The examiner's assessment of candidates will be based on: overall attention to sequence · composure · speed of performance.

### Instruments for neurologic examination

- Inelastic measuring tape
- Pen torch
- Tuning fork (256 Hz)
- Tendon hammer
- Cotton wool

### Higher function

**Consciousness** – conscious, lethargic, obtundation, stupor, coma.

**Intelligence & memory** – alertness for a younger child; simple arithmetic for an older child.

**Speech** – inability to speak (**aphasia**, due to damage to **Broca's area**) or inability to speak properly (**dysarthria**, due to damage to the articulation system).

**Orientation** – of time, place and person.

**Cerebral dominance** – left or right. Most people are left.

### Cranial nerves

|   | Nerve      |
|---|------------|
| 1 | Olfactory  |
| 2 | Optic      |
| 3 | Oculomotor |
| 4 | Trochlear  |
| 5 | Trigeminal |
| 6 | Abducent   |

|    | Nerve            |
|----|------------------|
| 7  | Facial           |
| 8  | Auditory         |
| 9  | Glossopharyngeal |
| 10 | Vagus            |
| 11 | Accessory        |
| 12 | Hypoglossal      |

### Cranial nerve function testing

- **Visual acuity, field of vision, colour vision & ocular fundi** – CN 2
- **Pupillary reactions** – CN 2, 3
- **Extraocular movements** – CN 3, 4, 6
- **Corneal reflexes and jaw movements** – CN 5
- **Facial movements** – CN 7
- **Hearing** – CN 8
- **Swallowing and rise of the palate** – CN 9, 10
- **Voice** – CN 10; **Speech** – CN 5, 7, 10, 12
- **Tongue movement** – CN 12

### Motor system examination

#### Tests to assess tone of the muscle

| Test                           | Normal | Hypotonia        | Hypertonia       |
|--------------------------------|--------|------------------|------------------|
| Palpation of muscles           | Normal | <b>Flabby</b>    | <b>Rigid</b>     |
| Posture of limb                | Normal | <b>Limp</b>      | <b>Stiff</b>     |
| Resistance to passive movement | Normal | <b>Decreased</b> | <b>Increased</b> |
| Range of passive movement      | Normal | <b>Increased</b> | <b>Decreased</b> |

Patients with either **spasticity or rigidity** might exhibit **opisthotonos** – defined as **severe hyperextension of the spine caused by hypertonia of the paraspinal muscles**.

Opisthotonos – severe hyperextension of the spine from hypertonia of the paraspinal muscles

#### Muscle power – graded 0 to 5

| Grade    | Movement                                                             |
|----------|----------------------------------------------------------------------|
| <b>0</b> | None                                                                 |
| <b>1</b> | Flickering movement                                                  |
| <b>2</b> | Movement possible after elimination of gravity (horizontal movement) |
| <b>3</b> | Movement possible against gravity but not against resistance         |
| <b>4</b> | Movement possible against gravity and against some resistance        |
| <b>5</b> | Normal power                                                         |

### Grading of reflexes



| Reflexes                                    | Grade            |
|---------------------------------------------|------------------|
| Absent reflexes                             | <b>0</b>         |
| Sluggish reflexes (seen with reinforcement) | <b>1+ (+)</b>    |
| Normal                                      | <b>2+ (++)</b>   |
| Brisk                                       | <b>3+ (+++)</b>  |
| Brisk with clonus                           | <b>4+ (++++)</b> |

## Special diagnostic procedures – lumbar puncture

**Lumbar puncture (LP) and cerebrospinal fluid (CSF) examination** is a clinical procedure in which a **needle is inserted into the spinal canal**, most commonly to **collect CSF for diagnostic testing**.

- LP is also known as **spinal tap**
- It is an **invasive procedure**, hence **requires consent**
- **Strict asepsis is required**
- It can also be for **therapeutic purposes**
- It has **indications and contraindications**

## Summary of requirements for LP procedure

1. Indications
2. Rule out contraindications
3. Vital signs
4. Pre-procedure concomitant RBS
5. Position
6. Needle insertion point / needle type
7. Asepsis
8. Pain control
9. Specimen collection

## Indications for lumbar puncture

- **Meningitis**
- **Encephalitis** (autoimmune, infectious)
- **Idiopathic intracranial hypertension** (previously referred to as pseudotumor cerebri)
- Often helpful in assessing **subarachnoid haemorrhage (SAH)**; **demyelinating, degenerative, and collagen vascular diseases**; and **intracranial neoplasms**

## Contraindications to lumbar puncture

1. **Suspected mass lesion of the brain**, especially in the **posterior fossa** or **above the tentorium** and causing **shift of the midline**
2. **Suspected mass lesion of the spinal cord**
3. **Symptoms and signs of impending cerebral herniation** in a child with probable meningitis
4. **Critical illness** (on rare occasions)
5. **Skin infection at the site** of the lumbar puncture
6. **Thrombocytopenia with a platelet count of  $< 20 \times 10^9 /L$**

## LP procedure

**Position** – lateral decubitus or seated position with the neck and legs flexed to enlarge the intervertebral spaces, while shoulders and hips are straight to prevent rotating the spine.

**Needle insertion point** – the appropriate interspace is **L3-L4 or L4-L5**, identified by drawing an imaginary line from the iliac crest downward perpendicular to the vertebral column.

**Asepsis** – the physician dons a **mask, gown, and sterile gloves**. The skin is thoroughly prepared with a cleansing agent, and sterile drapes applied.

**Pain control** – the skin and underlying tissues are anaesthetized by injecting a **local anaesthetic** (e.g. 1% lidocaine) at the time of the procedure, or by applying a **eutectic mixture of lidocaine and prilocaine (EMLA)** to the skin **30 minutes before** the procedure.

**Needle** – a **22-gauge, 1.5- to 3.0-inch, sharp, bevelled spinal needle with a properly fitting stylet**, introduced in the **midsagittal plane and directed slightly cephalad**.

## Investigations of the central nervous system

### Neuroradiologic procedures

**Skull X-rays** – have **limited diagnostic utility**. They can demonstrate **fractures, bony defects, intracranial calcifications, or indirect evidence of increased ICP**.

**Cranial ultrasonography** – the **imaging method of choice** for detecting **intracranial haemorrhage, periventricular leukomalacia, and hydrocephalus in infants with patent anterior fontanels**.

**Cranial CT** – a valuable diagnostic tool in the evaluation of **many neurologic emergencies**, as well as some non-emergent conditions.

**Brain MRI** – a **non-invasive** procedure that is well suited for detecting a variety of abnormalities, including those of the **posterior fossa and spinal cord**.

### Other neuroradiologic procedures:

- **Cranial CT angiography, MR angiography and MR venography**
- **Proton MR spectroscopy (MRS)**
- **Catheter angiography**
- **Positron emission tomography (PET)** – provides unique information on **brain metabolism and perfusion** by measuring **blood flow, oxygen uptake, and/or glucose consumption**

### Electroencephalography (EEG)

EEG is for the **diagnosis of epilepsy and brain dysfunction**.

### Evoked potentials

An **evoked potential** is an **electrical signal recorded from the CNS following the presentation of a specific visual, auditory, or sensory stimulus**.

### Genetic and metabolic testing

Children with **intellectual disability or developmental delay** are often evaluated with **metabolic and/or genetic testing**.

### **Suggested reading**

- Nelson Textbook of Paediatrics – Kliegman, Stanton, St. Geme, Schor and Behrman
- Textbook of Paediatrics and Child Health in a Tropical Region – Azubuike and Nkanginieme